

1. Network Layer Protocols

- IP (IPv4, IPv6)
- ICMP
- ARP
- RARP

2. Transport Layer Protocols

- TCP
- UDP
- SCTP

3. Application Layer Protocols

- HTTP/HTTPS
- FTP
- SMTP
- POP3
- IMAP
- DNS
- DHCP
- SNMP
- Telnet
- SSH
- NTP

4. Data Link Layer Protocols

- Ethernet
- PPP
- HDLC
- Wi-Fi (IEEE 802.11)

5. Security Protocols

- SSL/TLS

- IPSec
- Kerberos
- OAuth
- SAML

6. Routing Protocols

- RIP
- OSPF
- EIGRP
- BGP

7. File Transfer Protocols

- SFTP
- SCP
- TFTP

8. Wireless and IoT Protocols

- MQTT
- Zigbee
- Bluetooth
- NFC

9. Streaming and Multimedia Protocols

- RTP (Real-Time Protocol)
- RTSP (Real-Time Streaming Protocol)
- RTMP (Real-Time Messaging Protocol)
- SIP (Session Initiation Protocol)

10. Virtualization and Cloud Protocols

- VNC (Virtual Network Computing)
- RDP (Remote Desktop Protocol)
- OpenFlow
- VXLAN (Virtual Extensible LAN)

11. Wireless Communication Protocols

- LTE (Long-Term Evolution)
- GSM (Global System for Mobile Communications)
- CDMA (Code Division Multiple Access)
- 5G/4G/3G

12. File Sharing Protocols

- SMB/CIFS (Server Message Block/Common Internet File System)
- NFS (Network File System)
- AFP (Apple Filing Protocol)

13. Database Protocols

- JDBC (Java Database Connectivity)
- ODBC (Open Database Connectivity)
- MySQL Protocol

14. Authentication and Access Protocols

- LDAP (Lightweight Directory Access Protocol)
- RADIUS
- TACACS+
- PAM (Pluggable Authentication Module)

15. IoT and M2M Protocols

- CoAP (Constrained Application Protocol)
- AMQP (Advanced Message Queuing Protocol)
- LwM2M (Lightweight M2M)

16. Industrial Protocols

- Modbus
- OPC-UA (Open Platform Communications Unified Architecture)
- PROFIBUS (Process Field Bus)

17. Storage Protocols

- iSCSI (Internet Small Computer Systems Interface)
- Fibre Channel
- FCoE (Fibre Channel over Ethernet)

18. Cryptographic and Key Exchange Protocols

- AES (Advanced Encryption Standard)
- RSA (Rivest-Shamir-Adleman)
- Diffie-Hellman
- HMAC (Hash-Based Message Authentication Code)

19. Email Security Protocols

- SPF (Sender Policy Framework)
- DKIM (DomainKeys Identified Mail)
- DMARC (Domain-based Message Authentication, Reporting, and Conformance)

20. Legacy Protocols

- NetBIOS
- X.25
- Frame Relay

Common Protocols

- **20, 21:** FTP (File Transfer Protocol)
- **22:** SSH (Secure Shell)
- **23:** Telnet
- **25:** SMTP (Simple Mail Transfer Protocol)
- **53:** DNS (Domain Name System)
- **67, 68:** DHCP (Dynamic Host Configuration Protocol)
- **69:** TFTP (Trivial File Transfer Protocol)
- **80:** HTTP (Hypertext Transfer Protocol)

- **110:** POP3 (Post Office Protocol v3)
- **123:** NTP (Network Time Protocol)
- **143:** IMAP (Internet Message Access Protocol)
- **161, 162:** SNMP (Simple Network Management Protocol)
- **389:** LDAP (Lightweight Directory Access Protocol)
- **443:** HTTPS (HTTP Secure)

Specialized Protocols

- **514:** Syslog
- **631:** IPP (Internet Printing Protocol)

Registered Ports (1024–49151)

File Sharing and Remote Access

- **1080:** SOCKS Proxy
- **1433:** Microsoft SQL Server
- **1521:** Oracle Database
- **2049:** NFS (Network File System)
- **3306:** MySQL
- **3389:** RDP (Remote Desktop Protocol)
- **5900–5901:** VNC (Virtual Network Computing)

Streaming and Multimedia

- **554:** RTSP (Real-Time Streaming Protocol)
- **1935:** RTMP (Real-Time Messaging Protocol)

Dynamic/Private Ports (49152–65535)

These ports are often used for ephemeral communication and do not have fixed assignments.

Additional Common Ports

- **465:** SMTPS (SMTP Secure)
- **993:** IMAPS (IMAP Secure)
- **995:** POP3S (POP3 Secure)
- **5060, 5061:** SIP (Session Initiation Protocol)
- **8080:** Alternative HTTP

File Transfer Services

- **20:** FTP (Data Transfer)
- **21:** FTP (Control)
- **69:** TFTP (Trivial File Transfer Protocol)
- **115:** SFTP (Simple File Transfer Protocol)
- **989, 990:** FTPS (FTP Secure)

Web Services

- **80:** HTTP (Hypertext Transfer Protocol)
- **443:** HTTPS (HTTP Secure)
- **8080:** Alternative HTTP
- **8443:** Alternative HTTPS

Email Services

- **25:** SMTP (Simple Mail Transfer Protocol)
- **110:** POP3 (Post Office Protocol v3)
- **143:** IMAP (Internet Message Access Protocol)
- **465:** SMTPS (SMTP Secure)
- **587:** SMTP (Mail Submission)
- **993:** IMAPS (IMAP Secure)
- **995:** POP3S (POP3 Secure)

Remote Access

- **22:** SSH (Secure Shell)
- **23:** Telnet
- **3389:** RDP (Remote Desktop Protocol)
- **5900–5901:** VNC (Virtual Network Computing)
- **1494:** Citrix ICA

Domain Name Services

- **53:** DNS (Domain Name System)

Directory and Authentication Services

- **88:** Kerberos
- **389:** LDAP (Lightweight Directory Access Protocol)
- **636:** LDAPS (LDAP Secure)

Network Management Services

- **161, 162:** SNMP (Simple Network Management Protocol)
- **514:** Syslog

Database Services

- **1433:** Microsoft SQL Server
- **1521:** Oracle Database
- **3306:** MySQL
- **5432:** PostgreSQL

Media Streaming

- **554:** RTSP (Real-Time Streaming Protocol)

- **1935:** RTMP (Real-Time Messaging Protocol)
- **7070:** RealAudio/RealVideo

Messaging and Communication

- **5060, 5061:** SIP (Session Initiation Protocol)
- **6667:** IRC (Internet Relay Chat)

File Sharing

- **137–139:** NetBIOS
- **445:** SMB/CIFS (Server Message Block/Common Internet File System)
- **2049:** NFS (Network File System)

Other Services

- **123:** NTP (Network Time Protocol)
- **1812, 1813:** RADIUS (Remote Authentication Dial-In User Service)
- **3128:** Squid Proxy
- **8081:** Jenkins

Other Monitoring Services and Their Ports

- **Nagios:**
 - Port 5666: NRPE (Nagios Remote Plugin Executor)
 - Port 5667: NSCA (Nagios Service Check Acceptor)
- **Zabbix:**
 - Port 10050: Zabbix agent
 - Port 10051: Zabbix server
- **Prometheus:**
 - Port 9090: Prometheus server
 - Port 9100: Node Exporter
- **Grafana:**
 - Port 3000: Grafana web interface
- **Datadog:**

- Port 8125: DogStatsD (UDP)

1. Physical Layer (Layer 1)

- **Attacks:**
 - **Wiretapping:** Intercepting signals on physical cables or wireless signals.
 - **Signal Jamming:** Disrupting wireless communication by overwhelming the frequency.
 - **Hardware Tampering:** Physically modifying devices to manipulate their behavior.
 - **Device Theft:** Stealing network devices like routers, switches, or servers.
- **Mitigation:**
 - Use physical security measures (locks, surveillance cameras).
 - Shield cables and use secure wireless protocols.
 - Employ tamper-evident seals.

2. Data Link Layer (Layer 2)

- **Attacks:**
 - **MAC Spoofing:** Using a fake MAC address to impersonate another device.
 - **Switch Port Stealing (MAC Flooding):** Overloading a switch's MAC table to force it into broadcast mode.
 - **ARP Spoofing:** Sending fake ARP messages to associate the attacker's MAC with another IP.
 - **VLAN Hopping:** Gaining unauthorized access to different VLANs.
- **Mitigation:**
 - Enable port security on switches.
 - Use dynamic ARP inspection (DAI).
 - Disable unused ports and implement VLAN segmentation.

3. Network Layer (Layer 3)

- **Attacks:**
 - **IP Spoofing:** Sending packets with a fake source IP address.
 - **Route Injection:** Inserting malicious routes into routing tables (e.g., BGP hijacking).
 - **Ping of Death:** Sending oversized ICMP packets to crash a target system.
 - **Smurf Attack:** Exploiting ICMP broadcast requests to overwhelm a victim.
- **Mitigation:**
 - Use ingress/egress filtering to block spoofed packets.
 - Employ secure routing protocols (e.g., BGP with RPKI).
 - Configure firewalls to block malicious ICMP traffic.

4. Transport Layer (Layer 4)

- **Attacks:**
 - **SYN Flood:** Exploiting the TCP handshake by sending many incomplete SYN requests.
 - **UDP Flood:** Overloading a system with excessive UDP packets.
 - **Port Scanning:** Scanning for open ports to identify vulnerabilities.
 - **Reset (RST) Attacks:** Forcing TCP connections to reset.
- **Mitigation:**
 - Use SYN cookies and rate-limiting on firewalls.
 - Employ intrusion detection/prevention systems (IDS/IPS).
 - Close unused ports and use port-knocking techniques.

5. Session Layer (Layer 5)

- **Attacks:**
 - **Session Hijacking:** Taking over an active session by stealing session cookies or IDs.
 - **SSL/TLS Stripping:** Downgrading HTTPS connections to HTTP.
- **Mitigation:**
 - Use secure session management and encryption (HTTPS).
 - Implement HTTP Strict Transport Security (HSTS).
 - Use strong authentication methods.

6. Presentation Layer (Layer 6)

- **Attacks:**
 - **Man-in-the-Middle (MITM):** Intercepting and altering data during transmission.
 - **Data Injection:** Modifying or injecting data in the middle of a session.
 - **SSL/TLS Vulnerabilities:** Exploiting weak encryption algorithms or outdated SSL/TLS protocols.
- **Mitigation:**
 - Use strong encryption and up-to-date protocols (TLS 1.2/1.3).
 - Validate certificates to prevent MITM attacks.
 - Employ end-to-end encryption.

7. Application Layer (Layer 7)

- **Attacks:**
 - **SQL Injection:** Executing malicious SQL queries to access or modify databases.
 - **Cross-Site Scripting (XSS):** Injecting malicious scripts into web pages viewed by users.
 - **Cross-Site Request Forgery (CSRF):** Forcing users to execute unwanted actions on a web app.
 - **Distributed Denial of Service (DDoS):** Overwhelming a server with excessive traffic.
 - **Phishing:** Trick users into revealing sensitive information via fake websites or emails.
 - **DNS Spoofing/Poisoning:** Redirecting users to malicious websites.
- **Mitigation:**
 - Sanitize and validate input data to prevent injections.
 - Use Content Security Policy (CSP) to mitigate XSS.
 - Implement rate-limiting and Web Application Firewalls (WAFs).
 - Train users to recognize phishing attempts.

Summary Table

OSI Layer	Example Attacks
Physical (1)	Wiretapping, Signal Jamming
Data Link (2)	MAC Spoofing, ARP Spoofing, VLAN Hopping
Network (3)	IP Spoofing, Route Injection, Smurf Attack
Transport (4)	SYN Flood, UDP Flood, Port Scanning
Session (5)	Session Hijacking, SSL/TLS Stripping
Presentation (6)	MITM, Data Injection, Weak Encryption
Application (7)	SQL Injection, XSS, DNS Spoofing

Let me know if you'd like deeper insights on any of these attacks!

4o

You've hit the Free plan limit for GPT-4o.

Responses will use another model until your limit resets after 8:40 PM.

Get Plus

File and Directory Management

- **ls** – List directory contents.
 - Example: `ls -l` (long format), `ls -a` (include hidden files)
- **cd** – Change directory.
 - Example: `cd /path/to/directory`
- **pwd** – Print the current working directory.
- **mkdir** – Create a new directory.
 - Example: `mkdir myfolder`
- **rmdir** – Remove an empty directory.
- **rm** – Remove files or directories.
 - Example: `rm file.txt`, `rm -r folder` (recursive removal)
- **cp** – Copy files or directories.
 - Example: `cp file1.txt file2.txt`
- **mv** – Move or rename files or directories.
 - Example: `mv oldname.txt newname.txt`
- **touch** – Create an empty file or update the timestamp of an existing file.

- Example: `touch newfile.txt`
- `find` – Search for files in a directory hierarchy.
 - Example: `find /path -name "file.txt"`
- `locate` – Find files by name (faster than `find`).
 - Example: `locate file.txt`
- `cat` – Concatenate and display file content.
 - Example: `cat file.txt`
- `more/less` – View the contents of a file one page at a time.
 - Example: `less file.txt`
- `head` – Display the beginning of a file.
 - Example: `head -n 10 file.txt`
- `tail` – Display the end of a file.
 - Example: `tail -n 10 file.txt`, `tail -f logfile` (follow updates)

System Information

- `uname` – Display system information.
 - Example: `uname -a` (all details)
- `hostname` – Show or set the system's hostname.
- `uptime` – Show how long the system has been running.
- `top` – Display real-time system resource usage.
- `htop` – Interactive version of `top` (requires installation).
- `ps` – Show current running processes.
 - Example: `ps aux` (all processes), `ps -ef` (extended format)
- `df` – Display disk space usage.
 - Example: `df -h` (human-readable format)
- `du` – Display disk usage of files and directories.
 - Example: `du -sh /path/to/folder`
- `free` – Show memory usage.
 - Example: `free -h` (human-readable format)
- `vmstat` – Display system performance related to processes, memory, paging, block IO, etc.
- `lscpu` – Show CPU architecture information.
- `lsblk` – List information about block devices (disks).

User and Group Management

- `whoami` – Display the current logged-in user.
- `id` – Show user ID (UID) and group ID (GID) information.
- `useradd` – Add a new user.
 - Example: `useradd newuser`
- `usermod` – Modify a user account.
 - Example: `usermod -aG sudo user` (add user to sudo group)
- `passwd` – Change a user's password.
- `groupadd` – Add a new group.
- `groups` – Display the groups a user belongs to.
- `chown` – Change file owner and group.
 - Example: `chown user:group file.txt`
- `chmod` – Change file permissions.
 - Example: `chmod 755 file.txt` (read, write, execute for owner; read, execute for group and others)
- `chgrp` – Change group ownership of a file.

Package Management (Debian-based)

- `apt update` – Update package index.
- `apt upgrade` – Upgrade installed packages.
- `apt install` – Install a package.
 - Example: `apt install vim`
- `apt remove` – Remove a package.
 - Example: `apt remove vim`
- `apt search` – Search for a package.
- `dpkg -l` – List installed packages.
- `dpkg -i` – Install a package from a `.deb` file.

Networking

- `ping` – Send ICMP Echo Request to test network connectivity.
 - Example: `ping google.com`
- `ifconfig` – Display or configure network interfaces (deprecated, replaced by `ip`).

- **ip** – Show/manipulate network interfaces, routing, and devices.
 - Example: `ip addr show, ip link set eth0 up`
- **netstat** – Show network connections, routing tables, and interface statistics.
 - Example: `netstat -tuln` (list active listening ports)
- **ss** – Show socket statistics (alternative to netstat).
 - Example: `ss -tuln`
- **wget** – Download files from the web.
 - Example: `wget http://example.com/file.tar.gz`
- **curl** – Transfer data from or to a server.
 - Example: `curl -O http://example.com/file.tar.gz`
- **scp** – Securely copy files between hosts.
 - Example: `scp file.txt user@remote:/path/to/destination`
- **ssh** – Securely connect to a remote host.
 - Example: `ssh user@remotehost`
- **traceroute** – Trace the path packets take to a destination.
 - Example: `traceroute google.com`
- **nslookup** – Query DNS records.
 - Example: `nslookup google.com`
- **hostname -I** – Display the IP address of the system.

File Compression and Archiving

- **tar** – Archive and compress files.
 - Example: `tar -czvf archive.tar.gz files/` (create compressed tarball)
- **gzip** – Compress files.
 - Example: `gzip file.txt`
- **gunzip** – Decompress .gz files.
 - Example: `gunzip file.txt.gz`
- **zip** – Compress files into .zip archive.
 - Example: `zip archive.zip file1 file2`
- **unzip** – Extract files from a .zip archive.

Search and Text Processing

- **grep** – Search for patterns in a file.
 - Example: `grep 'pattern' file.txt`
- **sed** – Stream editor for text manipulation.
 - Example: `sed 's/old/new/' file.txt` (replace text)
- **awk** – Pattern scanning and processing language.
 - Example: `awk '{print $1}' file.txt` (print first column)
- **cut** – Remove sections from lines of files.
 - Example: `cut -d ',' -f 1 file.csv` (extract first field in CSV)
- **sort** – Sort lines of text files.
 - Example: `sort file.txt`
- **uniq** – Report or omit repeated lines.
 - Example: `uniq file.txt`
- **tee** – Read from standard input and write to both standard output and files.
 - Example: `echo "text" | tee file.txt`

Process Management

- **ps** – Display the current processes.
 - Example: `ps aux` (show all processes)
- **top** – Display real-time system resource usage.
- **kill** – Terminate a process.
 - Example: `kill PID`
- **killall** – Kill all processes by name.
 - Example: `killall firefox`
- **bg** – Resume a paused job in the background.
- **fg** – Bring a job to the foreground.
- **jobs** – List active jobs.

Disk Management

- **fdisk** – Partition table manipulator.
- **mount** – Mount a filesystem.
 - Example: `mount /dev/sda1 /mnt`

- **umount** – Unmount a filesystem.
 - Example: `umount /mnt`
- **lsblk** – List information about block devices.
- **parted** – Command-line partitioning tool.

System Shutdown and Reboot

- **shutdown** – Shutdown the system.
 - Example: `shutdown -h now` (halt immediately)
- **reboot** – Reboot the system.
- **halt** – Stop all processes and shutdown the system.
- **poweroff** – Turn off the system.

25 Unique Linux Commands

1. **ls** – List directory contents.
 - a. Example: `ls -l` (long format)
2. **cd** – Change directory.
 - a. Example: `cd /path/to/directory`
3. **pwd** – Print the current working directory.
4. **mkdir** – Create a new directory.
 - a. Example: `mkdir newfolder`
5. **rmdir** – Remove an empty directory.
6. **rm** – Remove files or directories.
 - a. Example: `rm -r folder/` (recursive)
7. **cp** – Copy files or directories.
 - a. Example: `cp file1.txt file2.txt`
8. **mv** – Move or rename files or directories.
 - a. Example: `mv oldname.txt newname.txt`
9. **cat** – Concatenate and display file content.
 - a. Example: `cat file.txt`
10. **touch** – Create an empty file or update a file's timestamp.
 - a. Example: `touch newfile.txt`
11. **find** – Search for files in a directory hierarchy.
 - a. Example: `find /path -name "file.txt"`

12. **locate** – Find files by name (uses a database).
 - a. Example: `locate file.txt`
13. **grep** – Search for patterns in files.
 - a. Example: `grep 'pattern' file.txt`
14. **head** – Display the first few lines of a file.
 - a. Example: `head -n 10 file.txt`
15. **tail** – Display the last few lines of a file.
 - a. Example: `tail -n 10 file.txt`
16. **df** – Display disk space usage.
 - a. Example: `df -h` (human-readable format)
17. **du** – Display disk usage of files and directories.
 - a. Example: `du -sh /path/to/folder`
18. **top** – Display real-time system resource usage.
 - a. Example: `top` (interactive)
19. **ps** – Show current running processes.
 - a. Example: `ps aux`
20. **kill** – Terminate a process by its PID.
 - a. Example: `kill 1234` (kill process with PID 1234)
21. **chmod** – Change file permissions.
 - a. Example: `chmod 755 file.txt`
22. **chown** – Change the ownership of a file.
 - a. Example: `chown user:group file.txt`
23. **ip** – Show or manipulate network interfaces.
 - a. Example: `ip addr show` (display IP addresses)
24. **wget** – Download files from the web.
 - a. Example: `wget http://example.com/file.tar.gz`
25. **tar** – Archive files.
 - a. Example: `tar -czvf archive.tar.gz folder/` (create a compressed archive)

These commands cover a wide range of areas such as file management, process management, networking, and system monitoring. You can explain how each command is used in a practical context, which will demonstrate your Linux proficiency to the interviewer.

4o mini

You said:

GIVE ME RARELY USED COMMANDS

ChatGPT said:

ChatGPT

Here's a list of **rarely used but powerful Linux commands** that might impress an interviewer with your advanced Linux knowledge. These commands cover niche functionalities and some specialized tools in the Linux ecosystem.

Rarely Used but Powerful Linux Commands

1. **n1** – Number lines of a file.
 - a. Example: `n1 file.txt` (numbers the lines of a text file)
2. **shred** – Securely delete a file by overwriting it.
 - a. Example: `shred -u file.txt` (overwrite and remove file)
3. **chroot** – Change the root directory for a process.
 - a. Example: `chroot /path/to/newroot /bin/bash` (run a command in a chroot environment)
4. **xargs** – Build and execute command lines from standard input.
 - a. Example: `find . -name "*.txt" | xargs wc -l` (count lines in all .txt files)
5. **tac** – Concatenate and display files in reverse.
 - a. Example: `tac file.txt` (prints file contents in reverse order)
6. **od** – Dump a file in various formats (octal, hex, ASCII, etc.).
 - a. Example: `od -c file.txt` (shows file in character format)
7. **tee** – Read from standard input and write to both standard output and files.
 - a. Example: `echo "Hello" | tee file.txt` (output to terminal and save to a file)
8. **stat** – Display detailed information about a file or file system.
 - a. Example: `stat file.txt` (show file metadata)
9. **watch** – Execute a program periodically and show the output.
 - a. Example: `watch ls -l` (run `ls -l` every 2 seconds)
10. **yes** – Output a string repeatedly until interrupted.
 - a. Example: `yes "Hello World"` (keeps printing "Hello World" indefinitely)
11. **time** – Measure the execution time of a command.
 - a. Example: `time ls` (shows the time taken by `ls` command to execute)

12. **basename** – Strip the directory and suffix from filenames.
 - a. Example: `basename /path/to/file.txt` (outputs `file.txt`)
13. **dirname** – Extract the directory path from a file.
 - a. Example: `dirname /path/to/file.txt` (outputs `/path/to`)
14. **cmp** – Compare two files byte by byte.
 - a. Example: `cmp file1.txt file2.txt` (shows the first difference between two files)
15. **fold** – Wrap text to fit in a specified width.
 - a. Example: `fold -w 50 file.txt` (wrap text at 50 characters)
16. **strace** – Trace system calls and signals.
 - a. Example: `strace ls` (shows system calls made by the `ls` command)
17. **lsof** – List open files and their corresponding processes.
 - a. Example: `lsof -i :80` (list processes using port 80)
18. **htop** – Interactive process viewer (better alternative to `top`).
 - a. Example: `htop` (real-time system process viewer with advanced options)
19. **unbuffer** – Disable output buffering for commands.
 - a. Example: `unbuffer command` (prevents command output from being buffered)
20. **ss** – Display socket statistics.
 - a. Example: `ss -tuln` (list active listening TCP/UDP ports)
21. **nc (Netcat)** – Network utility for reading/writing data across network connections.
 - a. Example: `nc -l 1234` (listen on port 1234 for incoming connections)
22. **dd** – Copy and convert files (used for disk cloning).
 - a. Example: `dd if=/dev/sda of=/dev/sdb bs=64K` (clone a disk)
23. **at** – Schedule a command to run once at a specific time.
 - a. Example: `at 14:00` (schedule a task to run at 2 PM)
24. **fuser** – Identify processes using a file or a socket.
 - a. Example: `fuser -v /path/to/file` (shows processes using the specified file)
25. **seq** – Generate a sequence of numbers.
 - a. Example: `seq 1 10` (outputs numbers from 1 to 10)

`tr` command in Linux stands for **translate**

Convert Lowercase to Uppercase: Convert all lowercase letters in a string to uppercase.

bash

Copy code

```
echo "hello world" | tr 'a-z' 'A-Z'
```


Routing Protocols Overview

1. RIP (Routing Information Protocol)

- **Type:** Distance Vector
- **Metric:** Hop Count (Max 15 hops, 16 is considered unreachable)
- **Convergence Time:** Slow (typically 30 seconds to update entire routing table)
- **Timer Information:**
 - **Update Timer:** Every 30 seconds
 - **Invalid Timer:** 180 seconds (if no updates, route is considered invalid)
 - **Holddown Timer:** 180 seconds (prevents routing loops)
 - **Flush Timer:** 240 seconds (route is flushed from the table after 240 seconds if still invalid)
- **Routing Table Size:** Limited (Only supports 15 hops)
- **Class:** IGP (Interior Gateway Protocol)
- **Protocols Supported:** IPv4 only (RIPng for IPv6)
- **Use Case:** Small, simple networks.

2. OSPF (Open Shortest Path First)

- **Type:** Link-State
- **Metric:** Cost (based on bandwidth)
- **Convergence Time:** Fast (typically under 1 second for stable networks)
- **Timer Information:**
 - **Hello Interval:** 10 seconds (for point-to-point links), 30 seconds (broadcast)
 - **Dead Interval:** 40 seconds (before declaring a neighbor down)
 - **LSA Refresh Interval:** 30 minutes (for LSRs)
 - **LSA Age:** 60 minutes
- **Routing Table Size:** Scalable for large networks (supports multiple routes per destination)
- **Class:** IGP (Interior Gateway Protocol)
- **Protocols Supported:** IPv4, IPv6
- **Use Case:** Medium to large-scale enterprise networks.

3. EIGRP (Enhanced Interior Gateway Routing Protocol)

- **Type:** Hybrid (combines aspects of Distance Vector and Link-State)
- **Metric:** Bandwidth, Delay, Load, MTU (composite metric)
- **Convergence Time:** Fast (typically under 1 second in most cases)
- **Timer Information:**
 - **Hello Interval:** 5 seconds (for point-to-point), 60 seconds (for broadcast/multicast)
 - **Hold Time:** 15 seconds (before declaring a neighbor down)
 - **Update Timer:** Every 5 seconds (partial updates)
 - **Query Timer:** 3 minutes
- **Routing Table Size:** Flexible, optimized for larger networks
- **Class:** IGP (Interior Gateway Protocol)
- **Protocols Supported:** IPv4, IPv6
- **Use Case:** Larger and more complex networks, especially when there are different types of traffic.

4. BGP (Border Gateway Protocol)

- **Type:** Path Vector
- **Metric:** AS Path, Prefix Length, Next Hop, Local Preference, MED (Multi-Exit Discriminator)
- **Convergence Time:** Slow (can take minutes depending on network size)
- **Timer Information:**
 - **Keepalive Timer:** 60 seconds
 - **Hold Timer:** 180 seconds (if no hello messages are received, session is terminated)
 - **Update Interval:** BGP does not send periodic updates; it sends updates only when there is a change in the routing table.
- **Routing Table Size:** Can handle large-scale networks (global internet routing)
- **Class:** EGP (Exterior Gateway Protocol)
- **Protocols Supported:** IPv4, IPv6, VPNv4 (MPLS)
- **Use Case:** Inter-domain (Internet) routing, between ISPs, and large-scale networks.

5. IS-IS (Intermediate System to Intermediate System)

- **Type:** Link-State
- **Metric:** Cost (administered by the administrator)
- **Convergence Time:** Moderate (usually under 1 second, depending on network size)
- **Timer Information:**
 - **Hello Interval:** 10 seconds (default)
 - **Dead Interval:** 40 seconds (neighbor is declared down)
 - **LSP Refresh:** 15 minutes
- **Routing Table Size:** Suitable for large-scale networks
- **Class:** IGP (Interior Gateway Protocol)
- **Protocols Supported:** IPv4, IPv6
- **Use Case:** Large-scale enterprise networks, service provider networks.

Comparison of Routing Protocols: Key Metrics

Routing Protocol	Type	Metric	Convergence Time	Hello Interval	Dead Interval	Max Hop Count	Use Case
RIP	Distance Vector	Hop Count	Slow (30-60 sec)	30 sec	180 sec	15 hops	Small networks, simple routing
OSPF	Link-State	Cost (Bandwidth)	Fast (under 1 sec)	10 sec	40 sec	Unlimited	Large enterprise networks
EIGRP	Hybrid	Bandwidth, Delay, Load, MTU	Fast (under 1 sec)	5 sec	15 sec	Unlimited	Larger, complex networks
BGP	Path Vector	AS Path, Prefix Length, MED	Slow (minutes)	60 sec	180 sec	Unlimited	Inter-domain routing (Internet)
IS-IS	Link-State	Cost (administered)	Moderate (under 1 sec)	10 sec	40 sec	Unlimited	Service providers, large networks

Other Important Numbers:

- **RIP:**
 - Maximum number of hops: **15** (16 is unreachable)
 - Default update interval: **30 seconds**
 - Default invalid timer: **180 seconds**
- **OSPF:**
 - Default hello interval: **10 seconds** (Point-to-Point links)
 - Default dead interval: **40 seconds**
 - LSA age: **60 minutes**
 - LSR refresh interval: **30 minutes**
- **EIGRP:**
 - Hello interval: **5 seconds** (Point-to-Point), **60 seconds** (Broadcast/Multicast)
 - Hold time: **15 seconds**
 - Update interval: **5 seconds**
 - Query interval: **3 minutes**
- **BGP:**
 - Keepalive interval: **60 seconds**
 - Hold time: **180 seconds**
 - Update interval: **Event-driven (no periodic updates)**
- **IS-IS:**
 - Hello interval: **10 seconds**
 - Dead interval: **40 seconds**
 - LSP refresh interval: **15 minutes**

Interior Gateway Protocols (IGPs)

IGPs are used within an **Autonomous System (AS)**.

Distance Vector IGPs:

- **RIP (Routing Information Protocol)**
 - **Version 1 (RIPv1):** The first version of RIP.
 - **Version 2 (RIPv2):** An updated version of RIP that supports classless routing.
 - **Maximum Number of Hops:** 15 (16 is unreachable).

Link-State IGPs:

- **OSPF (Open Shortest Path First)**
 - **OSPFv2:** Version 2 supports IPv4.
 - **OSPFv3:** Version 3 supports IPv6.
 - **Convergence Time:** Fast convergence with low overhead.
- **IS-IS (Intermediate System to Intermediate System)**
 - **Used:** Primarily in large-scale networks and service providers.
 - **Convergence Time:** Similar to OSPF but uses a different algorithm.

Hybrid IGP:

- **EIGRP (Enhanced Interior Gateway Routing Protocol)**
 - **Type:** A hybrid protocol (combines distance vector and link-state).
 - **Convergence Time:** Fast convergence.
 - **Metric:** Based on bandwidth, delay, load, and reliability.

2. Exterior Gateway Protocols (EGPs)

EGPs are used to exchange routing information between different Autonomous Systems (ASes).

Path Vector Protocol:

- **BGP (Border Gateway Protocol)**
 - **BGPv4:** The most common version used today for routing across the internet.
 - **Uses:** AS path, prefix length, next-hop, and various attributes for decision-making.

Summary of Routing Protocols:

Protocol	Type	Metric	Used for
RIP	Distance Vector	Hop Count	Small Networks
OSPF	Link-State	Cost (Bandwidth)	Large Enterprise Networks
EIGRP	Hybrid	Bandwidth, Delay	Large/Complex Networks

BGP	Path Vector	AS Path	Inter-domain Routing
IS-IS	Link-State	Cost	Service Providers, Large Networks